School of Engineering and Applied Science

Lecture Scribe: Lecture 22 (CSE 400)

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1) List of Topics Covered

1. **Markov's Inequality Recap**
   - Application to income distribution problems.
2. **Chebyshev's Inequality Key Idea**
   - Utilizing mean and variance to control deviations.
3. **Intuition and Geometry**
   - Understanding tail regions and large deviations from the mean.
4. **Proof of Chebyshev's Inequality**
   - Step-by-step derivation using Markov's Inequality.
5. **Comparative Analysis**
   - Chebyshev vs. Markov bounds in a coin-flipping scenario.
6. **Real-Life Applications**
   - Estimating student score distributions and grade cutoffs.

2) Explanation of Topics Covered

1. **Markov's Inequality**

   This inequality provides an upper bound on the probability that a non-negative random variable X exceeds a certain value a. It is primarily a "right-tail" bound and requires only the expected value ($E[X]$).

2. **Chebyshev's Inequality**

   Unlike Markov's, Chebyshev's inequality uses both the mean (μ) and the variance ($Var(X)$) of a distribution. It controls the probability that a random variable deviates from its mean by more than a specific distance a.

3. **Intuition**

   Geometrically, Chebyshev's inequality bounds the area in the "tail regions" far from the mean. As the spread (variance) around the mean decreases, the probability of finding values far from the center also decreases.

4. **Key Insight**

   One major advantage of Chebyshev's inequality is that its bound often decreases as the sample size n increases (as seen in the coin-flip example), whereas Markov's bound may remain constant. This makes it a significantly tighter and more useful tool for large datasets.

3) List of Definitions and Theorems

1. **Definition: Markov's Inequality**
   Let X be a non-negative random variable ($X \geq 0$) with a known expected value $E[X]$. For any $a > 0$:
   $$P(X \geq a) \leq \frac{E[X]}{a}$$

2. **Theorem: Chebyshev's Inequality**
   Let X be a random variable with mean μ and variance $Var(X)$. For any $a > 0$, the probability that X deviates from μ by at least a is:
   $$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

3. **Proof of Chebyshev's Inequality**
   The proof relies on applying Markov's Inequality to the squared deviation of X from its mean:

   1. **Identity**: Note that the event $|X - \mu| \geq a$ is equivalent to the event $(X - \mu)^2 \geq a^2$.
   2. **Application**: Define a new non-negative random variable $Y = (X - \mu)^2$. Apply Markov's Inequality to Y with threshold a^2:
      $$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$
   3. **Substitution**: Since $E[Y] = E[(X - \mu)^2] = Var(X)$, we substitute this into the inequality:
      $$P((X - \mu)^2 \geq a^2) \leq \frac{Var(X)}{a^2}$$
   4. **Result**: Thus, $P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$.

4) Important Examples

Example 1: Income Distribution (Markov)
Question: In a city, the average annual income is $40,000. Use Markov's inequality to find an upper bound on the probability that a randomly selected person earns $120,000 or more.

Solution:

- Define $X \geq 0$ as income, with $E[X] = 40,000$.
- Apply $P(X \geq a) \leq E[X]/a$ with $a = 120,000$.
- $P(X \geq 120,000) \leq \frac{40{,}000}{120{,}000} = \frac{1}{3}$.

Note: Markov cannot bound the probability of earning less than $10,000 because it only handles the right tail.

Example 2: Coin Flipping (Chebyshev vs. Markov)
Question: Flip a fair coin n times. Let X be the number of heads. Bound $P(X \geq \frac{3n}{4})$.

Solution:

- Setup: $X \sim Binomial(n, 1/2)$, $\mu = n/2$, $Var(X) = n/4$.
- Using Chebyshev: $P(X \geq \frac{3n}{4}) = \frac{1}{2}P(|X - \frac{n}{2}| \geq \frac{n}{4})$.
- Apply bound: $\leq \frac{1}{2} \cdot \frac{n/4}{(n/4)^2} = \frac{2}{n}$.

Insight: The bound $\frac{2}{n}$ decreases as n increases, which is much better than the constant bound provided by Markov's inequality.

Example 3: Exam Scores (Real-Life Application)
Question: Mean exam score is 70 marks and standard deviation is 10. What fraction of students score below 40 or above 100?

Solution:

- Identify deviation: The scores 40 and 100 both deviate from the mean (70) by $a = 30$.
- Apply Chebyshev: $P(|X - 70| \geq 30) \leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$.

At most 1/9 of students are far from the mean.

Example 4: Exam Grades with Variance (Exercise 5.8b)
Question: The average grade is 70% and the standard deviation is 5%. The "A" cutoff is 90%. Use Chebyshev's inequality to find an upper bound on the fraction of "A" students.

Solution:

- Identify parameters: $\mu = 70$, $\sigma = 5 \Rightarrow Var(X) = 25$.

- Identify deviation: An "A" student ($X \geq 90$) deviates from the mean by at least 20 ($|X-70| \geq 20$).
- Apply bound: $P(X \geq 90) \leq P(|X-70| \geq 20) \leq \frac{25}{20^2}$.
- Calculation: $\frac{25}{400} = \frac{1}{16} = 0.0625$ or 6.25%.

5) List of Important Formulas

Formula / Concept	**Result**
Markov's Inequality (Requires $X \geq 0$)	$P(X \geq a) \leq \frac{E[X]}{a}$
Chebyshev's Inequality (Uses mean and variance)	$P(\lvert X-\mu\rvert \geq a) \leq \frac{Var(X)}{a^2}$
Variance Calculation (Fundamental to Chebyshev proof)	$Var(X) = E[(X-\mu)^2]$
Binomial Mean (Used in coin-flip example)	$\mu = np$
Binomial Variance (Used in coin-flip example)	$Var(X) = npq$